Lab 7: Fuzzing for Input Validation Bugs

Vivaan J Atharva | PES1UG24AM071 | PES_SE_B_2026 (Section B)
Repository: github.com/Vivaan-Atharva/L6_L7_09_bookstore_trade_in (folder *Lab 7*)

Step 1: Install dependencies

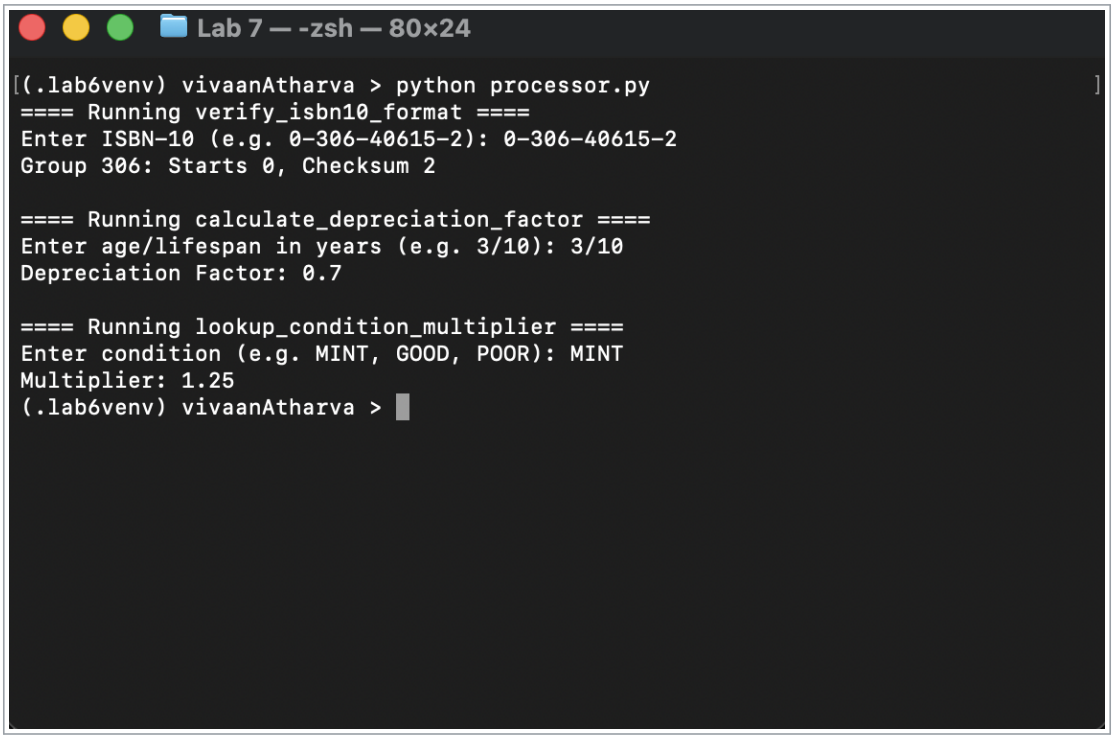
```
pip install hypothesis pytest
```

Hypothesis 6.170.15 and pytest 9.2.0 on Python 3.12.4 (macOS).

Step 2: Run the buggy application

```
python processor.py
```

With well-formed inputs (the examples from the prompts) the original script produces the expected output.

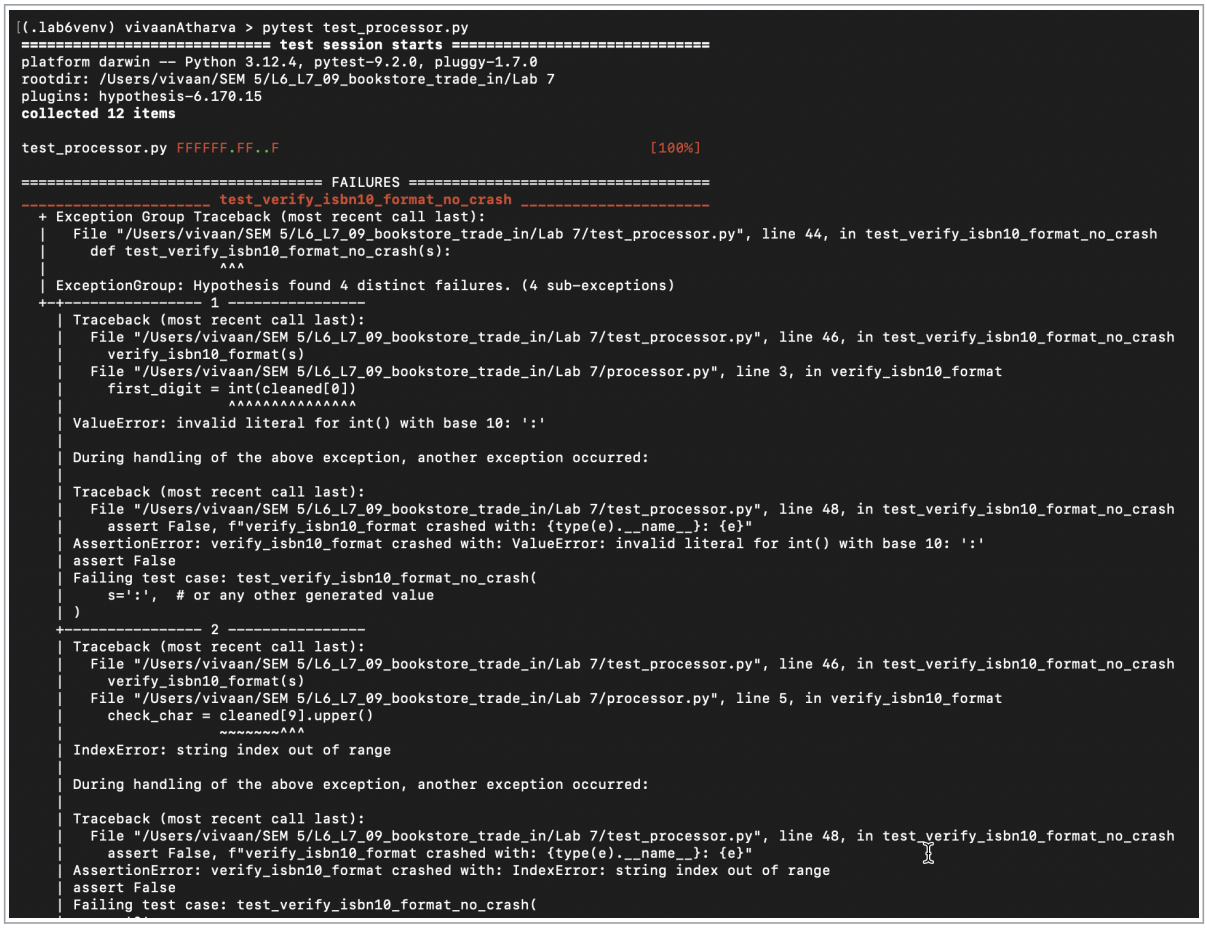


Steps 3 and 4: Fuzz tests written with Hypothesis, run against the buggy script

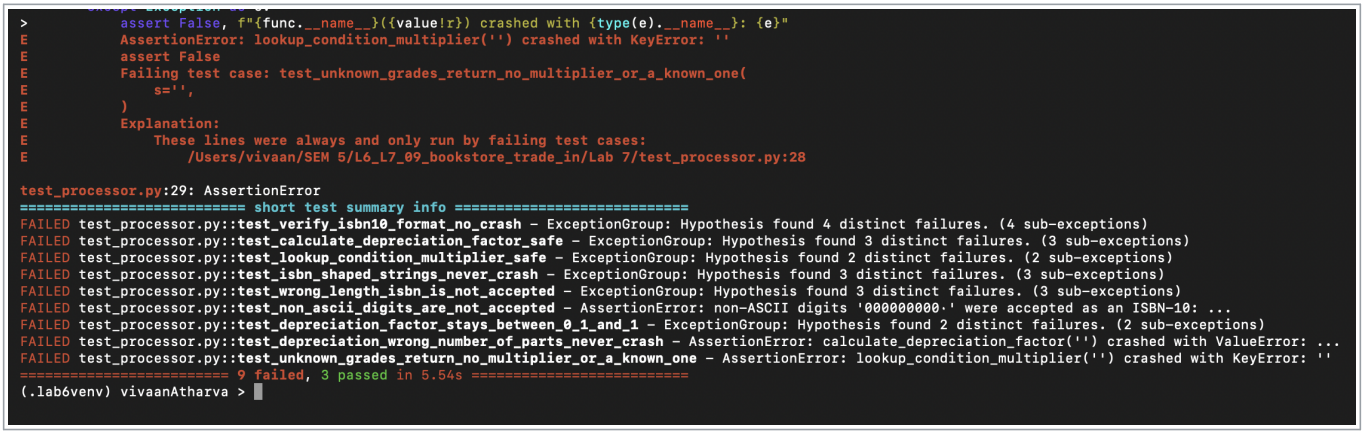
```
pytest test_processor.py
```

test_processor.py completes the three starter tests (random text or None must not crash any function) with try/except, and adds nine targeted properties: ISBN-shaped strings, wrong-length and non-ASCII-digit ISBNs, valid ISBNs with correct checksums, numeric age/lifespan pairs including 0, negatives, nan and inf, monotonic depreciation, and known grades in any case and padding. Against the original script **9 of 12 tests fail**.

Start of the run: Hypothesis finds 4 distinct crashes in verify_isbn10_format alone (ValueError on ':' and IndexError on '0' and '').



End of the run: short test summary, **9 failed, 3 passed**.



Bugs found by fuzzing and how they were fixed

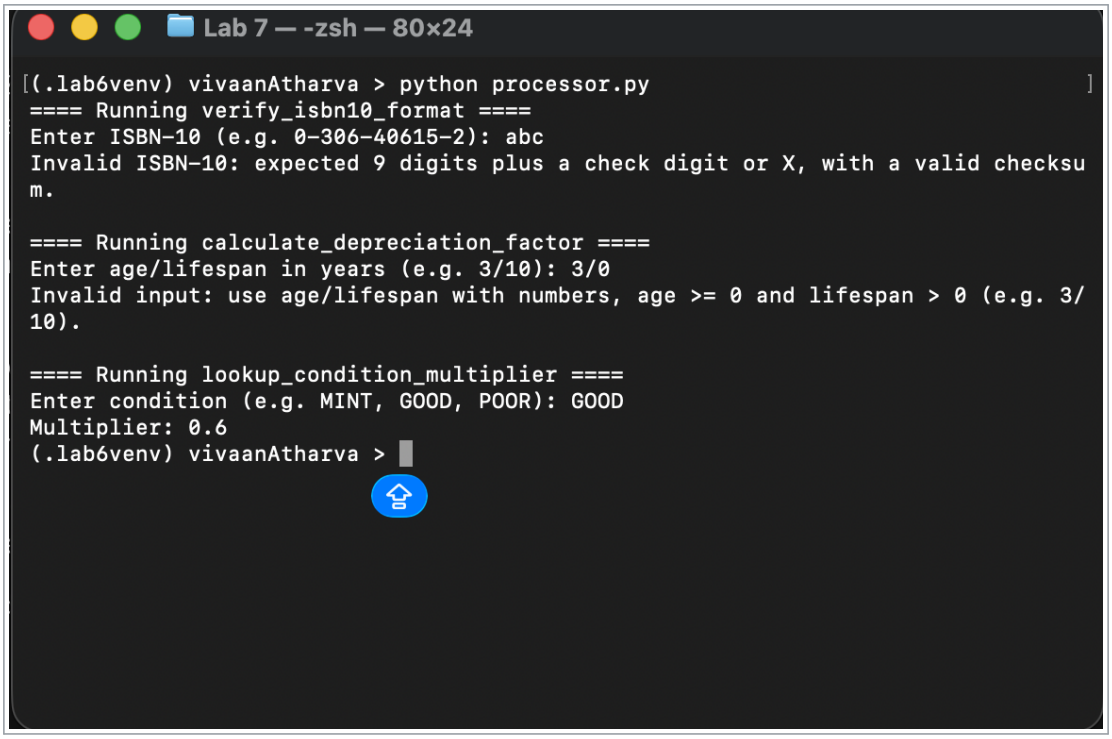
Each function now validates its input and returns None for anything invalid instead of raising; main() prints a clear message in that case. Valid inputs give exactly the same output as before.

Function	Input	Original behaviour	Fix
`verify_isbn10_format`	`'' , '0'`	IndexError	length must be exactly 10 after removing hyphens/spaces
	`':'`	ValueError from int()	first nine characters must be ASCII digits
	`None`	AttributeError	non-string input returns None
	`nine '0's + an Arabic-Indic zero`	accepted as valid	ASCII digits only, check digit or X, checksum verified
	`'03064061520' (11 digits)`	accepted as valid	length check
`calculate_depreciation_factor`	`'' , '3' , '1/2/3'`	ValueError (unpack)	exactly one '/' required
	`'abc/10'`	ValueError from float()	non-numeric parts return None
	`'3/0'`	ZeroDivisionError	lifespan must be > 0
	`'-1/1'`	factor 2.0 (above 1.0)	age must be >= 0; result clamped to [0.1, 1.0]
	`None`	AttributeError	non-string input returns None; nan/inf rejected
`lookup_condition_multiplier`	`'' , 'EXCELLENT'`	KeyError	unknown grade returns None
	`None`	AttributeError	non-string input returns None

Step 5: Run the fixed application

```
python processor.py
```

Invalid inputs ('abc' and '3/0') now produce clear messages instead of crashing, and a valid grade ('GOOD') still returns its multiplier.



Step 6: Re-run the fuzz tests against the fixed script

```
pytest test_processor.py
```

All 12 tests pass.

